

Introduction to Special Relativity

Chapter 3: Mass-Energy Equivalence and Relativistic Momentum

Study Notes Series

1 Redefining Momentum

In Newtonian mechanics, momentum is simply $p = mv$. But this definition, combined with the Lorentz transformation from Chapter 1, no longer conserves momentum consistently across reference frames at high speeds. Special relativity fixes this with a modified definition:

$$p = \gamma mv, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (1)$$

As $v \rightarrow c$, the factor $\gamma \rightarrow \infty$, so an object with nonzero mass would need infinite momentum, and therefore infinite energy, to reach the speed of light. This is the mathematical reason massive objects can approach but never reach c .

2 The Most Famous Equation in Physics

Einstein showed that the total energy of a moving object is

$$E = \gamma mc^2. \quad (2)$$

When the object is at rest ($v = 0$, so $\gamma = 1$), this reduces to the famous rest-energy relation:

$$E_0 = mc^2. \quad (3)$$

This equation states that mass itself is a form of energy. A small amount of mass corresponds to an enormous amount of energy, because it is multiplied by $c^2 \approx 9 \times 10^{16} \text{ m}^2/\text{s}^2$. This is why nuclear reactions, which convert a tiny fraction of mass into energy, release far more energy per kilogram of fuel than chemical reactions like combustion, which merely rearrange chemical bonds without converting mass.

Worked example

Nuclear fission of one kilogram of uranium-235 converts roughly 0.1% of its mass into energy. Using $E = mc^2$ with $m = 0.001 \text{ kg}$:

$$E = (0.001 \text{ kg}) \times (3 \times 10^8 \text{ m/s})^2 = 9 \times 10^{13} \text{ J}, \quad (4)$$

comparable to the energy released by a large chemical explosion many thousands of times over.

3 The Energy-Momentum Relation

Combining the expressions for E and p eliminates γ and v , yielding a relation that holds for every particle, including massless ones like photons:

$$E^2 = (pc)^2 + (mc^2)^2. \quad (5)$$

For a photon, $m = 0$, so the relation simplifies to $E = pc$: photons carry momentum despite having no rest mass, which is the principle behind solar sails and radiation pressure.

4 Why This Matters Beyond Physics Class

Mass-energy equivalence and relativistic kinematics are not just theoretical curiosities. GPS satellites must correct their onboard clocks for both time dilation (from their orbital speed) and gravitational time dilation (a topic in general relativity) or they would accumulate positioning errors of several kilometers per day. Particle accelerators, such as the Large Hadron Collider, are designed entirely around relativistic momentum and energy, since the particles inside routinely reach speeds above $0.999c$.

Quick check

If a particle's rest energy is $E_0 = 500$ MeV and it moves with $\gamma = 2$, what is its total energy? (Answer: $E = \gamma E_0 = 1000$ MeV.)